

SKILLS

Languages: Rust, Python, TypeScript, Java, C++, Gleam

ML/AI: Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK

Frameworks: React, Flask, Litestar, FastAPI

Databases: LanceDB, MySQL, MongoDB, DynamoDB

Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

EMPLOYMENT

LexisNexis Legal & Professional

Raleigh, NC

Senior Data Scientist I

May 2024 - Present

- Technical lead for an agentic legal drafting system; led its migration from RAG and continue to own development.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Built a contrastive-learning filter to remove unrelated legal passages from search results; median filtering latency of 12ms across 40 queries.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- **Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University

Raleigh, NC

PhD Student

Aug 2019 - Jul 2024

- **State-of-the-art hyper-parameter optimization:** Proposed a novel hyper-parameter optimization method that outperforms prior SOTA and is 36.4% faster.
- **Better, faster deep learning for SE:** Improved defect prediction by up to 123% (F-1 score), code smell detection by up to 30% (AUC), issue lifetime prediction by up to 76% (accuracy), automated microservice partitioning by up to 285% (modularity) compared to prior SOTA.
- **Semi-supervised learning:** Achieved state-of-the-art results (up to 100% improvement in AUC) on static code warnings analysis using 10% of the labels.
- **Teaching:** TA for 830 students over 5 semesters across CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

RECENT PUBLICATIONS

See full list on [Google Scholar](#).

1. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering*, 51(6).
2. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM*, 66(2), 75-81.
3. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering*, 49(4), 2856-2872.
4. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications*, 219, 119673.
5. **Yedida, R.**, Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.

FUNDING

\$5,000, Google Cloud Academic Research Grant, Feb 2022

SERVICE TO PROFESSION

Contributing Member, Python Software Foundation (Mar 2021 - Present)

Guest Editor, IEEE Software SI: The Impact of AI on Productivity and Code 2025; ASEj SI: Replications and Negative Results (RENE) 2025; EMSE SI: Replications and Negative Results (RENE) 2025

Workshop Co-Chair, Intelligent Software Engineering (ASE '25, ICST '26); Replications and Negative Results (ASE '24)

PC Member, AAAI 2025-2027; FSE 2026-2027; ICSE 2026-2027; EDM 2026; ASE Industry Track 2026; AI Foundation Models and Software Engineering (FORGE) 2024; ICSME Artifact Evaluation Track, 2021-2023; ASE Artifact Evaluation Track, 2022; International Conference on Modeling, Machine Learning, and Astronomy (MMLA), 2019

Reviewer, TOSEM 2026; TSE 2025-2026; NeurIPS 2023, 2025; EMSE 2021, 2025; ASEj 2025; ICML 2024-2025; ICLR 2024-2025; NCAA 2023-2025; TMLR 2024; Neural Processing Letters 2023-2024; Artificial Intelligence Review 2023; J. Big Data, 2023; ASE 2023; IEEE SSCI 2020

HONORS

Distinguished Reviewer, ICSE 2026 Mar 2026

Google Developer Expert - ML and GCP Mar 2025 - Present
A global network of experts recognized by Google for deep technical expertise, community leadership, and mentorship.

Google Cloud Champion Innovator - Cloud AI/ML Oct 2022 - Mar 2025
A Googler-nominated global network of 500+ technical experts in Google Cloud; merged into the Google Developer Experts program in 2025.

Google Cloud Research Innovator Feb 2022
Research Innovators are a global community of researchers driving scientific breakthroughs with Google Cloud.

RELEVANT PROJECTS

pi-permission-system (maintained fork) May 2026 - Present
TypeScript, Node.js [GitHub](#)
Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero Oct 2024 - Present
Rust, SQLite, LanceDB [GitHub](#) :: [crates.io](#)
Built an academic question-answering and semantic search system over a user's Zotero library. Wrote a custom PDF parser for academic papers (median throughput $\approx$ 2,400 pages/s across 11 PDFs; M4 Pro) and a RAG library with support for multiple providers and tool calling.

RAISE Aug 2020 - Jun 2025
Python, Keras [GitHub](#) :: [PyPI](#)
Sole developer of an ML utilities Python package used by multiple research groups for reproducible experiments. Downloaded 74k+ times on PyPI.

Google/Meta Data Mining Feb 2021 - May 2021
Python, Keras [GitHub](#)
Mined Google Takeout and Meta data exports to predict which Amazon best-sellers to advertise to a user (DistilGPT-2, 0.6 F-1 score).

Novel Drug Repurposing Hypotheses Oct 2019 - Feb 2020
Python, PyTorch [GitHub](#)
Identified novel drug repurposing hypotheses by text mining transcripts of an NPR medical radio show, and verified results using a knowledge graph.

EDUCATION

North Carolina State University Raleigh, NC
Ph.D. Computer Science Aug 2019 - Jul 2024

- Advisor: Dr. Tim Menzies
- Dissertation: [Guidelines for the Application of Neural Technologies in Software Analytics \(or: How to Do More with Less in SE\)](#)

PES University Bangalore, India
B.E. Computer Science – Advisor: Dr. Snehanshu Saha Aug 2015 - May 2019